

Week 1 - Lecture 2 - Problem: The Collatz Sequence

Note: *You only need to read and understand the problem before the lecture. You are not required to produce a programming solution.*

ILO

- **KU1:** Grasp the relation between source code, the interpreter, and the machine.
- **KU3:** Design algorithms to solve simple programming problems.
- **CS1:** Structure small programs by the use of concepts such as iterations, functions, modules, classes, and methods.
- **CS3:** Form readable, descriptive and well-documented program code.
- **CS6:** Build basic interactive programs with text-based (and graphical) user interfaces.
- **CS8:** Use programming tools such as text editor, command line interface, and IDE (integrated development environment).

Objectives

After completing this problem, you should be able to:

1. Explain how Python source code is executed by the interpreter.
2. Translate the even and odd properties into an algorithm.
3. Use variables, arithmetic expressions, and an `if/else` statement to calculate the next value in the sequence.
4. Use a loop to generate values until the sequence reaches `1`.
5. Define and call a function.
6. Read input from the user and display the required output.
7. Write readable code with descriptive names, suitable formatting, and comments.
8. Run and check the program in a Python development environment using the provided example.

Background

The Collatz sequence is defined as follows for any positive integer n :

- If n is even, divide it by 2.
- If n is odd, multiply it by 3 and add 1.

Repeat this process until the value becomes 1.

For example, starting with 6 produces the sequence:

$6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

Task

Write a program that:

1. Reads a positive integer from the user.

2. Uses a function to generate the Collatz sequence.
3. Prints each value in the sequence.
4. Prints the total number of steps required to reach 1.

Example Run

```
Enter a positive integer: 6
```

```
6  
3  
10  
5  
16  
8  
4  
2  
1
```

```
Number of steps: 8
```

Requirements

- Use at least one function.
- Use a loop to generate the sequence.
- Assume the user enters a positive integer. ``